Van-Arkel method for refining zirconium/titanium:

This method is based on the thermal decomposition of metal compounds which lead to the formation of pure metals. Titanium and zirconium can be purified using this method. For example, the impure titanium metal is heated in an evacuated vessel with iodine at a temperature of 550 K to form the volatile titanium tetra-iodide.(TiI_4). The impurities are left behind, as they do not react with iodine.

$$Ti\ (s) + 2I_2\ (s) \xrightarrow[\Delta]{550K} TiI_4\ (vapour)$$

The volatile titanium tetraiodide vapour is passed over a tungsten filament at a temperature around 1800 K. The titanium tetraiodide is decomposed and pure titanium is deposited on the filament. The iodine is reused.

$$TiI_4\ (vapour) \xrightarrow[\Delta]{1800\ K} Ti\ (s) + 2I_2\ (s)$$

1.7 Applications of metals

1.7.1 Applications of Al

Aluminium is the most abundant metal and is a good conductor of electricity and heat. It also resists corrosion. The following are some of its applications.

- Many heat exchangers/sinks and our day to day cooking vessels are made of aluminium.
- It is used as wraps (aluminium foils) and is used in packing materials for food items,
- Aluminium is not very strong, However , its alloys with copper, manganese, magnesium and silicon are light weight and strong and they are used in design of aeroplanes and other forms of transport.
- As Aluminium shows high resistance to corrosion, it is used in the design of chemical reactors, medical equipments,refrigeration units and gas pipelines.
- Aluminium is a good electrical conductor and cheap, hence used in electrical overhead electric cables with steel core for strength.

1.7.2 Applications of Zn

- Metallic zinc is used in **galvanising** metals such as iron and steel structures to protect them from rusting and corrosion.
- Zinc is also used to produce die-castings in the automobile, electrical and hardware industries
- Zinc oxide is used in the manufacture of many products such as paints, rubber, cosmetics,

XII U1 Metallurgy - Jerald.indd 17 16-12-2022 18:53:03